MUNDRA ULTRA MEGA POWER PLANT



TATA

TATA POWER

Presented By

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Overview

Part of the Ultra Mega Power Project (UMPP) initiative by the Government of India to address the nation's growing power demand. Tata Power won the project through a competitive bidding process in 2006 by **quoting a levelized tariff of ₹2.26 per kWh.**

1. **Ownership and Capacity:** Developed by subsidiary of Tata called Coastal Gujarat Power Limited (CGPL), a wholly-owned subsidiary of Tata Power, with a **capacity of 4,000 MW.**

2. **Technology and Fuel:** Uses **supercritical boiler technology** for higher efficiency and reduced emissions, primarily powered by imported Indonesian coal.

3. **Power Distribution:** Supplies power to five states — **Gujarat, Maharashtra, Punjab, Haryana, and Rajasthan** — through **long-term power purchase agreements (PPAs)**.

4. **Financial and Operational Challenges:** Supported by a ₹ 2,025 Crore loan from the Asian Development Bank (ADB) but faced financial difficulties due to rising coal prices, impacting project viability.

Timeline Of The Project

2006: Tata Power won the project through a competitive bidding process.

2007: Financial closure was achieved, and construction began.

March 2012: The first unit of the power plant was commissioned.

March 2013: The entire 4,000 MW capacity (5 units of 800 MW each) became fully operational.

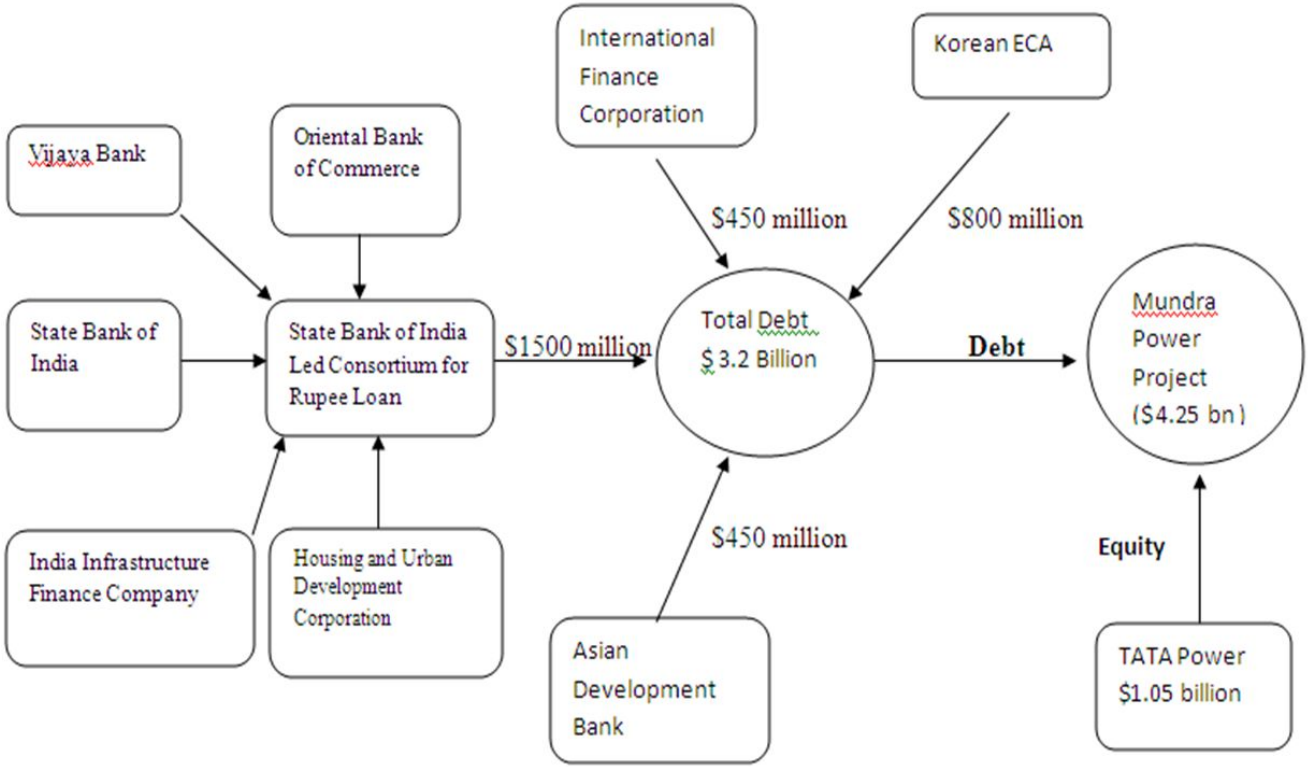
Financial Structure

- The Mundra UMPP follows a **Build-Own-Operate (BOO) model** within the PPP framework
- Tata Power through its subsidiary Coastal Gujarat Power Limited (CGPL) — is **responsible for financing, constructing, operating, and maintaining** the project.
- State Government signed **long-term Power Purchase Agreements (PPAs) for 25 years**, providing assured revenue and market access.

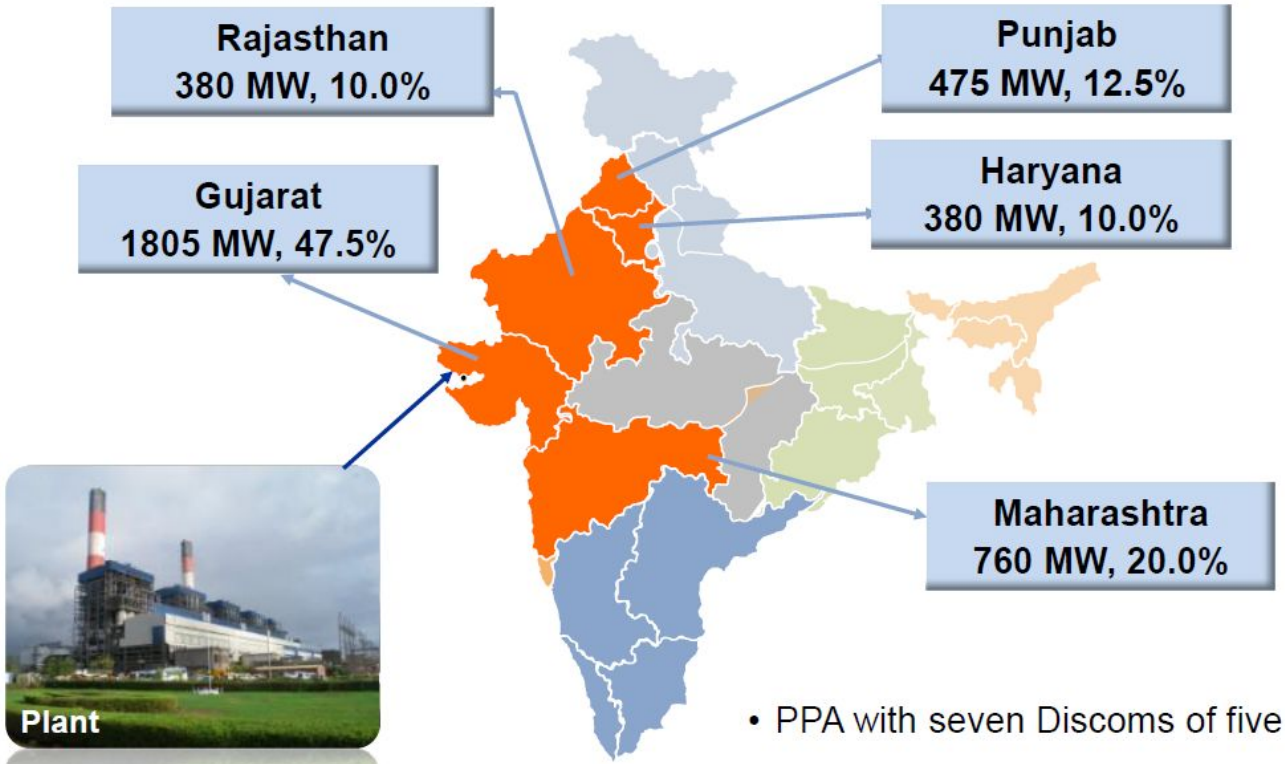
	RRP Amount (₹ crore)	Actual Amount (₹ crore)	Share Total
Equity	4,543	8,812	44.4%
Debt	13,631	11,050	55.6%
Total Sources of Funds	**18,174**	**19,866**	**100%**

- There was a **9.3% increase in project cost** compared to the RRP estimate.

Transaction Structure



State-wise Share



- PPA with seven Discoms of five States
- Lead Procurer - GUVNL

Tariff Agreement

- Tata Power won the Mundra UMPP bid with a levelized tariff of ₹2.26 per unit.
- Tariff breakdown:
  - Fuel charge: ₹0.86 per unit (45% escalable, 55% non-escalable)
  - Capacity charge: ₹0.88 per unit
  - Transportation and handling: ₹0.52 per unit
- Non-escalable fuel charge (55%) was fixed in dollar terms.
- Intended for stability against currency fluctuations but posed high risk if coal prices increased.

Problems With Tariff Agreement

The project encountered substantial challenges when Indonesia implemented new regulations in 2011 that mandated coal exporters to sell coal at market benchmark prices. This regulation effectively nullified TPC’s long-term coal supply contracts, which were previously negotiated at lower fixed rates. As a result, the coal price jumped from the $43 per ton assumed in TPC's original bid to $74 per ton by June 2012. This sharp increase in coal costs created significant financial stress for the project

Revenue Generation

1.Power Sale Revenue: The primary revenue source was electricity sales under 25-year Power Purchase Agreements (PPAs) with Gujarat, Haryana, Maharashtra, Punjab, and Rajasthan at a levelized tariff of ₹2.26 per unit.

2.Coal Mining Operations: Tata Power's 30% stake in Indonesian coal companies (PT Arutmin Indonesia and PT Kaltim Prima Coal) generated profits, intended to offset Mundra's coal cost risks. However, these profits couldn't fully cover Mundra's financial losses.

3.Clean Development Mechanism (CDM) Credits: Tata Power intended to earn CDM credits for reduced emissions, but registration was rejected as the project couldn't prove "additionality."

4.Internal Financial Support: To manage cash flow deficits, Tata Power provided additional equity, subordinate loans, and cross-subsidized Mundra's losses using profits from its other businesses.

5.Potential Tariff Compensation: Tata Power sought compensatory tariff relief to recover increased coal costs, but legal challenges delayed this potential revenue.

Key Risk & Allocation

1. **Coal Price Risk**

• Risk Description: Reliance on imported Indonesian coal with 55% of fuel charge fixed as non-escalable exposed the project to price volatility. Costs surged from $43 to $74 per ton in 2011 due to regulation changes.

• Risk Allocation: Tata Power secured long-term supply contracts and acquired stakes in Indonesian coal companies, but the new regulations made fixed-price contracts ineffective.

2. **Tariff Structure Risk**

• Risk Description: Non-escalable fuel charges limited tariff adjustments during price hikes.

• Risk Allocation: Despite seeking relief from CERC, courts ruled against tariff adjustments under the PPA.

Key Risk & Allocation

3. Currency Risk

- Risk Description: Fuel and transportation costs in US dollars led to currency risk.
- Risk Allocation: Partial hedging was attempted, but it proved insufficient.

4. Regulatory and Legal Risk

- Risk Description: Indonesian regulation changes increased fuel costs.
- Risk Allocation: Exemption requests and legal attempts for tariff adjustments failed.

Key Risk & Allocation

5. Operational and Performance Risk

- Risk Description: Blending high- and low-grade coal for cost management reduced efficiency.
- Risk Allocation: Despite challenges, plant availability was kept above 80%, meeting PPA standards.

6. Environmental and Social Risk

- Risk Description: Local fisherfolk opposed the project over marine ecological impact.
- Risk Allocation: Tata Power implemented a Remedial Action Plan for community support.

Expected Project Financials

Average DSCR	2.27
NPV	23810.44 (in Cr.)
Project IRR	11.48%
Equity IRR	17.96%
Margin of Safety	98.72%

Project Financials

Parameter	Value
TATA Power D/E	0.66
TATA Power Equity/Capital	0.2
Unlevered Beta	0.17
Coastal Gujarat Levered Beta	0.68
Market Risk Premium	7.00%
Risk-Free Rate	7.30%
Cost of Equity	12.06%
Weighted Average Cost of Debt	11.47%
WACC	9.04%

Profitability of Project

- UMPPs in India have often faced profitability issues due to factors such as fuel supply challenges, land acquisition delays, and unfavorable tariff structures. While some projects have achieved high operational efficiency, the overall financial viability of UMPPs has been inconsistent, influenced by a combination of operational hurdles and policy-related challenges.

- As of in 2024 the UMPP has show profit after almost a decade of Inconsistent revenue.

Forecasting the Project Future

For now this project has not been profitable as expect, but with the multiple upcoming Project in Gujarat like -

- **Tata Electronics** is establishing a display semiconductor manufacturing unit in Dholera, Gujarat, in partnership with Taiwan's PSMC and Himax Technologies
- **Kaynes Semicon Pvt Ltd** received approval to set up a semiconductor manufacturing unit in Sanand, Gujarat, with an investment of ₹3,300 crore
- **Micron Technology** is constructing an assembly and test facility in Gujarat, with Phase 1 including 500,000 square feet of cleanroom space, expected to become operational in late 2024.
- **Tata Aircraft Complex** in Vadodara, Gujarat, is India's first private military aircraft plant, established in collaboration with Airbus Spain to produce the Airbus C-295 transport military aircraft.
- **Maruti Suzuki** plans to produce its first electric vehicle at its plant in Gujarat, with production set to begin in spring 2025.

With this upcoming project the demand of electricity with be emerging suddenly and it is safe to say that in future UMPP is be profitable.

Risks During the Construction Phase

1.Technical Risks

Engineering & Design Challenges: Large-scale supercritical technology implementation required precision in design and execution.

Issues related to boiler efficiency, cooling systems, and ash handling could cause delays.

Supply Chain Disruptions: Delays in the procurement of turbines, generators, and boilers due to global supply chain issues. Heavy reliance on imported machinery led to logistical delays.

Workforce & Labor Management: Shortage of skilled workers in specialized fields like supercritical technology and high-voltage transmission.

2.Financial & Economic Risks

Cost Overruns: Higher-than-expected costs in materials, labor, and contractor charges.

Currency Exchange Rate Volatility: The project involved importing coal and machinery, making it vulnerable to currency fluctuations.

Financing & Investment Risks: The project relied on international financing (including from the World Bank's International Finance Corporation (IFC)), which imposed additional compliance conditions.

3.Environmental Risks

Impact on Coastal Ecosystem: Disruptions to marine life due to the intake and discharge of water for cooling.

Air & Water Pollution Risks: Coal combustion produces emissions such as SO_2, NOx, and CO_2, posing regulatory risks. Ash disposal and water contamination concerns.

Control Strategies for Risk Mitigation

1.Engineering & Construction Controls

Strict Quality Control Measures: Tata Power implemented project management software and real-time monitoring of construction.

Alternative Sourcing: Diversified sourcing strategies to reduce dependence on specific suppliers.

Skilled Workforce Training: Training programs for engineers and workers in supercritical technology were conducted.

2.Financial & Economic Controls

Hedging Strategies: Currency fluctuations were managed using forex hedging and financial instruments.

Project Financing Diversification: Funding from a mix of domestic and international financial institutions reduced dependency on a single lender.

Cost Optimization Measures: Efficient procurement and vendor negotiations helped in cost control.

3.Environmental & Social Controls

Environmental Impact Assessment (EIA) and Mitigation Measures: Adoption of electrostatic precipitators (ESPs), flue gas desulfurization (FGD) units, and efficient water treatment plants.

Corporate Social Responsibility (CSR) Initiatives: Tata Power initiated fishermen livelihood programs, healthcare centers, and local employment generation to address social concerns.

Legal Dispute Management: A dedicated legal team was set up to handle environmental and regulatory litigations.

Legal & Regulatory Compliance

Key Legal & Regulatory Bodies Involved

- Ministry of Environment, Forest and Climate Change (MoEFCC) – Approval for environmental clearance.
- Central Electricity Regulatory Commission (CERC) – Tariff regulation.
- Gujarat Pollution Control Board (GPCB) – Air, water, and solid waste regulations.
- International Finance Corporation (IFC) Guidelines – Due to external funding.
- Environmental Protection Act, 1986 – Ensured emissions & pollution controls.
- Coastal Regulation Zone (CRZ) Act – Since the plant is near the coast.
- The Air (Prevention and Control of Pollution) Act, 1981 – Emission norms compliance.

Legal Challenges Faced

Environmental Litigation: Activists and local communities filed cases citing pollution and displacement issues.
A lawsuit against IFC claimed inadequate environmental and social impact assessments.
Tariff Disputes: Tata Power sought tariff revisions due to increased coal costs (Indonesian coal pricing change).
CERC allowed partial compensation, but legal battles continued.

Impact of Legal Compliance

Positive Impacts

Increased Credibility & Funding Access: Adhering to IFC guidelines helped in securing international financing.
Improved Environmental Performance: Use of advanced pollution control technologies ensured regulatory compliance.
Better Social Relations: CSR programs and legal compliance helped in reducing public opposition.

Negative Impacts

Project Delays & Cost Escalation: Environmental clearances and legal disputes delayed project completion.
Higher costs due to compliance with additional environmental mitigation measures.
Operational Challenges: Compliance with new pollution norms required upgrading infrastructure (e.g., installation of FGD units).
Tariff Uncertainty & Financial Stress: Regulatory delays in approving tariff hikes affected revenue and profitability.

Cost-Benefit Analysis (CBA)

Cost Factors	Benefit Factors
Project Cost: ₹ 19,866 Cr	Electricity Generation: 4,000 MW power supply to five states
Coal Import & Fuel Cost: Fluctuating coal prices impacted operational expenses	Energy Security: Reduced power deficit in Gujarat, Maharashtra, Rajasthan, Haryana, and Punjab
Environmental Mitigation Cost: Air pollution controls (FGD, ESP), water treatment, and waste management.	Economic Growth: Industrial development, employment generation, and GDP boost
Resettlement & Compensation Costs: Rehabilitation of displaced communities	Lower Tariff for Consumers: Competitive pricing helped households and industries
Legal & Regulatory Compliance Costs: Tariff disputes, environmental litigation, and regulatory approvals	Infrastructure Development: Boosted roads, ports, and regional connectivity

Key Stakeholders & Their Interests

Stakeholder	Impact & Concerns
Government of India	✅ Energy security, economic growth, power availability
State Governments (Gujarat, Maharashtra, Rajasthan, Haryana, Punjab)	✅ Industrial and commercial benefits from power availability
Tata Power (Project Developer)	❌ Financial stress due to coal price fluctuations and tariff disputes
Local Communities & Fishermen	❌ Loss of livelihood, environmental impact, displacement
Regulatory Bodies (CERC, MoEFCC, IFC)	✅ Compliance enforcement, but legal battles delayed project outcomes
Investors & Lenders (IFC, Banks, Equity Holders)	❌ Increased risk due to financial uncertainties and operational costs
Industries & Businesses	✅ Reliable power supply supported economic growth

Social Impact – Key Challenges

Key Social Challenges

Displacement & Loss of Livelihoods
Fishermen and local villagers lost access to coastal areas due to plant operations.
Compensation and rehabilitation efforts were insufficient, leading to protests.

Environmental Impact on Communities
Marine biodiversity was affected due to thermal discharge and water pollution.
Air pollution from coal combustion affected local health.

Legal & Community Backlash
NGOs and activists challenged the project in Indian courts and international tribunals.
The World Bank's IFC was sued for failing to properly assess environmental risks.

Employment Generation vs. Displacement The project created jobs, but most opportunities were temporary.
Skilled labor was outsourced, leaving locals with minimal employment benefits.

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